MAHMOUD ELFIL

Junior Software Engineer | Full-Stack, Backend & Internal Tools

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SUMMARY

Junior software engineer in Berlin with industry experience building backend services, full-stack dashboards, desktop automation tools, Android/iOS workflows, and production-oriented internal systems. Hands-on with Python, TypeScript, React/Next.js, FastAPI, Node.js, Kotlin, Swift, Linux, Docker, CI/CD, REST APIs, and cloud/serverless tooling.

EXPERIENCE

Resistine GmbH - Desktop and Android Developer, Network Security | Berlin, Germany | Dec 2025 - Present
- Built a cross-platform Python/Qt tool that automates Wazuh agent installation, configuration, service setup, and reporting to the Wazuh server.
- Packaged and deployed Windows/Linux builds with Inno Setup, bundled dependencies, configuration files, and system-service setup.
- Developed Android modules in Kotlin for Wi-Fi/app risk scoring, evil-twin detection, WireGuard VPN integration, network checks, and Wazuh reporting.

IAV GmbH - Software Engineering Working Student | Berlin, Germany | Oct 2024 - Nov 2025
- Developed Python/FastAPI backend services for image uploads, dashboard data delivery, and customer authentication in software-solution workflows.
- Built approximately 20-50 React/Next.js dashboard pages and reusable components for customer engineers and internal managers.
- Developed and maintained a Swift iOS app for accessibility-focused image capture, quality filtering, preprocessing, and retake workflows.
- Maintained YOLOv7/YOLOv9 preprocessing pipelines for thousands of images and approximately 20-30 GB of automotive/accessibility-focused data.

SELECTED PROJECTS

TariffGuard - Serverless EV Charging Tariff Audit System | Python, AWS Lambda, API Gateway, DynamoDB, SQS, EventBridge, Cognito, Terraform, React/Vite
- Built an EV charging session validation platform with idempotent ingest, historical tariff lookup, Decimal-based price calculation, suspicious-session flags, auditable DynamoDB records, and an operations dashboard.
- Designed AWS/Terraform infrastructure with Cognito login, JWT-protected API Gateway routes, Python Lambda workers, SQS/DLQ processing, EventBridge audits, CloudWatch logs, and a React/Vite dashboard.

Planora - Academic Timetabling Platform | Python, OR-Tools CP-SAT, PyQt6, React/Vite, SQLite, Docker, Caddy
- Built a 12-week timetabling platform with CP-SAT feasibility solving, local-search improvement, configurable constraints, snapshots, imports, schedule comparison, and DOCX/CSV/ICS/PDF exports.
- Added a Python API server and React/Vite web client with authenticated workspaces, async solver jobs, health/readiness endpoints, SQLite persistence, audit logs, CI checks, and Docker/Caddy deployment.

TypeShift Station - Cloudflare Typing Platform | Next.js 15, React 19, TypeScript, Cloudflare Workers, D1, R2, Turnstile
- Built a Cloudflare-first typing platform with solo modes, custom dictionaries, daily challenges, seasonal ladders, sessions, replay sharing, live race rooms, tournaments, duel queues, webhooks, and legal pages.
- Implemented a Workers-native API with D1 persistence, R2 cache/backups, Turnstile abuse protection, token sessions, Worker deploy scripts, smoke checks, CI, and nightly database backups.

Anubis ARG Platform | Next.js 16, React 19, MongoDB, Docker
- Built a full-stack browser-riddle platform with sign-in, session/progress tracking, leaderboard persistence, local profile-image storage, hand-authored MDX/JSON puzzle levels, and Docker-ready deployment.

EDUCATION

German International University, Berlin - B.Sc. Networking Engineering | Expected July 2026 | GPA: 1.09 / 1.0 (A)

Selected coursework: Data Structures & Algorithms, Computer Organization & System Programming, Communication Networks, Internet, Communication Theory, Discrete Math & Numerical Methods

SKILLS

Backend / Full-Stack: Python, FastAPI, Node.js, Express, REST APIs, TypeScript, React, Next.js, Vite, PostgreSQL, MongoDB, SQLite, authentication
Cloud / DevOps: AWS Lambda, API Gateway, DynamoDB, SQS, EventBridge, Cognito, Terraform, Cloudflare Workers, D1, R2, Docker, Caddy, GitHub Actions, GitLab CI/CD
Mobile / Data / Desktop: Kotlin, Android, Swift, iOS, PyQt6, OR-Tools CP-SAT, YOLOv7/v9, OpenCV, Pandas, NumPy
Systems / Networking: Linux, Windows services, TCP/IP, WireGuard, Wazuh, Wireshark, Nmap | Languages: English, Arabic, German conversational